MECHANICAL PRESENTATION GUIDELINES

ROUND 1:PRESENTATION:

The objective of this task is to evaluate your technical understanding, research capability, and design approach towards converting our existing electric BAJA buggy into an autonomous vehicle.

Please go through the following general instructions carefully before starting your work

GENERAL INSTRUCTIONS:

Regardless of the specific problem statement you choose, every participant is expected to:

1. Understand the Existing Vehicle Systems
a) Study the current electric buggy design used in the SAE eBAJA competition.
b) Gain a clear understanding of the hydraulic braking system and throttle control design implemented in the vehicle.

2) Incorporate Safety and Reliability Principles
a) Always incorporate how manual override works for your proposed design
b) Design your proposed subsystem with redundant sensors and fail-safe mechanisms to ensure reliable operation.
c) Refer to relevant safety standards and guidelines (ISO, IATF, SAE, etc.) applicable to your subsystem.

3) Consider Environmental and Protection Factors
a) Ensure proper IP (Ingress Protection) rating when selecting electronic components
b) Include failsafe measures to handle system or power failures safely.

4) Demonstrate Research and Design Thinking
a) Show evidence of independent research, design reasoning, and practical feasibility.
b) Use **sketches, simulations, or block diagrams** to support your concept.
c) **BONUS**: show your interest by doing research on new developments like Regenerative braking, Adaptive Cruise Control, ADAS.

5)The simulation and design wherever applicable on top of critical thinking and logical reasoning will be considered as an important criterion for evaluation.

PROBLEM STATEMENTS:

PS1: Actuator Selection

Part A — Brake-by-Wire (BBW)

Design a simple actuator system to generate **2000 N force at the brake master cylinder** with a bore diameter (19.05mm)

- Explain how you will control the actuator.
- **Show how and where you will mount the actuator on the Baja buggy.**
- The mechanism should be capable of generating 2000N input force at the brake
- master cylinder.
- Required braking force and actuator torque/speed calculations, response time analysis
- Comparison of different mechanisms (including the different types of motors, actuator
- drive mechanism, and feedback control)

Part B — Steer-by-Wire (SBW)

Design a motorized steering system capable of providing **6–7 Nm torque** at the steering column.

- Select a suitable motor and reduction mechanism.
- Explain how you will control left/right steering.
- Add a method to limit steering angle.
- **Show how and where you will mount the system on the Baja buggy.**
- **Give a method steering angle from 30 degree to 35 degree**

PS2: Manual Override And Sensors integration

Part A — BBW

Design a BBW system that allows the driver to brake normally if the actuator fails and a basic sensor system for autonomous braking.

- Suggest a mechanical/electrical disengagement mechanism.
- Explain how manual braking is restored.
- **Show where the mechanism will be mounted on the Baja.**
- Select suitable sensors for brake pressure, wheel speed and vehicle deceleration.
- Explain how the controller uses these signals

Part B — SBW

Design an SBW system that allows the driver to **take manual control immediately** when autonomous steering is switched off.

- Select a suitable disengagement mechanism.
- Explain how the motor will be isolated.
- **Show how the mechanism will be mounted to the steering column/chassis.**
- **Give a method steering angle from 30 degree to 35 degree**
- **Explain how will you engage the motor and steering column**
- Select a suitable steering-angle sensor/encoder.
- Explain how steering position will be measured